INCH-POUND
MIL-PRF-49137/6D
22 April 2004
SUPERSEDING
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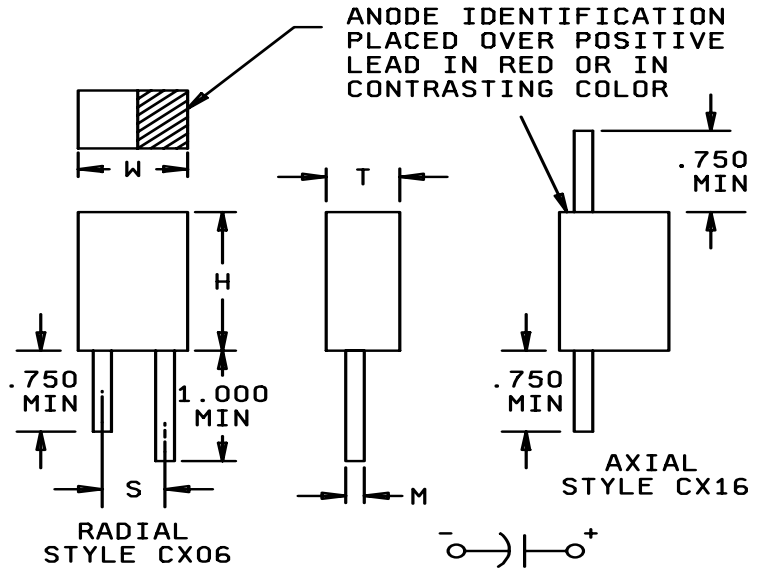
9 October 1997

PERFORMANCE SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC (SOLID ELECTROLYTE), TANTALUM, POLAR, NONHERMETICALLY SEALED, STYLES CX06 AND CX16

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-49137.



CIRCUIT DIAGRAM

FIGURE 1. Styles CX06 and CX16 capacitors.

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Case size	Dimensions				
	T	W	H	M	S
	Max	Max	Max	±.002	
A	.040	.050	.100	.007	.030 ±.015
B	.040	.070	.125	.010	.050 ±.015
C	.070	.120	.165	.010	.100 ±.020
D	.075	.185	.225	.010	.150 ±.020
E	.110	.220	.290	.016	.180 ±.025
F	.130	.230	.310	.016	.200 ±.025
G	.150	.375	.475	.016	.300 ±.025

Inches	mm	Inches	mm	Inches	mm	Inches	mm
.002	0.05	.040	1.02	.130	3.30	.230	5.84
.007	0.18	.050	1.27	.150	3.81	.290	7.37
.010	0.25	.070	1.78	.165	4.19	.300	7.62
.015	0.38	.075	1.90	.180	4.57	.310	7.87
.016	0.41	.100	2.54	.185	4.70	.375	9.52
.020	0.51	.110	2.79	.200	5.08	.475	12.06
.025	0.64	.120	3.05	.220	5.59	.750	19.05
.030	0.76	.125	3.18	.225	5.72	1.000	25.40

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The positive lead on CX06 shall be at least .125 inch (3.18 mm) longer than the negative lead.

FIGURE 1. Styles CX06 and CX16 capacitors - Continued.

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TABLE I. Styles CX06 and CX16 characteristics.

Type designation 1/	DC rated voltage (85°C)	Nominal capacitance	Dissipation factor (max)	DC leakage (max) 25°C	Case size
	Volts	µF	%	µA	
CX-6A474-	2	.47	10	.5	A
CX-6A225-	2	2.2	10	.5	B
CX-6A106-	2	10.0	10	.5	C
CX-6B155-	3	1.5	10	.5	B
CX-6B685-	3	6.8	10	.5	C
CX-6B226-	3	22.0	10	1.0	D
CX-6B227-	3	220.0	15	9.0	G
CX-6C334-	4	.33	10	.5	A
CX-6C105-	4	1.0	8	.5	B
CX-6C475-	4	4.7	8	.5	C
CX-6C156-	4	15.0	8	1.0	D
CX-6C476-	4	47.0	8	2.0	E
CX-6C686-	4	68.0	8	3.0	F
CX-6D224-	6	.22	10	.5	A
CX-6D684-	6	.68	6	.5	B
CX-6D335-	6	3.3	6	.5	C
CX-6D106-	6	10.0	6	1.0	D
CX-6D336-	6	33.0	6	2.0	E
CX-6D476-	6	47.0	6	3.0	F
CX-6D157-	6	150.0	10	9.0	G
CX-6F154-	10	.15	10	.5	A
CX-6F474-	10	.47	6	.5	B
CX-6F225-	10	2.2	6	.5	C
CX-6F685-	10	6.8	6	1.0	D
CX-6F226-	10	22.0	6	2.0	E
CX-6F336-	10	33.0	6	3.0	F
CX-6F107-	10	100.0	8	9.0	G
CX-6H104-	15	.10	10	.5	A
CX-6H334-	15	.33	6	.5	B
CX-6H155-	15	1.5	6	.5	C
CX-6H156-	15	15.0	6	2.0	E
CX-6H226-	15	22.0	6	3.0	F
CX-6H686-	15	68.0	8	9.0	G
CX-6J104-	20	.10	6	.5	B
CX-6J154-	20	.15	6	.5	B
CX-6J224-	20	.22	6	.5	B
CX-6J105-	20	1.0	6	.5	C
CX-6J335-	20	3.3	6	1.0	D
CX-6J475-	20	4.7	6	1.0	D
CX-6J106	20	10.0	6	2.0	E
CX-6J156-	20	15.0	6	3.0	F
CX-6J476-	20	47.0	8	9.0	G
CX-6K684-	25	.68	6	.5	C
CX-6K225-	25	2.2	6	1.0	D
CX-6K685-	25	6.8	6	2.0	E
CX-6K106-	25	10.0	6	3.0	F
CX-6K336-	25	33.0	6	9.0.	G

See footnote at end of table.

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TABLE I. Styles CX06 and CX16 characteristics. Continued.

Type Designation 1/	DC rated voltage (85°C)	Nominal capacitance	Dissipation factor (max)	DC leakage (max) 25°C	Case size
CX-6M104-	35	.10	6	.5	C
CX-6M154-	35	.15	6	.5	C
CX-6M224-	35	.22	6	.5	C
CX-6M334-	35	.33	6	.5	C
CX-6M474-	35	.47	6	.5	C
CX-6M684-	35	.68	6	1.0	D
CX-6M105-	35	1.0	6	1.0	D
CX-6M155-	35	1.5	6	1.0	D
CX-6M225-	35	2.2	6	2.0	E
CX-6M335-	35	3.3	6	2.0	E
CX-6M475-	35	4.7	6	2.0	E
CX-6M685-	35	6.8	6	3.0	F
CX-6M106-	35	10.0	6	9.0	G
CX-6M156-	35	15.0	6	9.0	G
CX-6M226-	35	22.0	6	9.0	G

1/ Complete type designation will include additional symbols to indicate style and capacitance tolerance.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Case: Polyester sleeve and epoxy end filled or plastic encapsulated.

Leads: Solder-coated metal.

DC rated voltage: See table I.

Operating temperature: -55° to +85°C.

DC leakage (DCL): See table I.

Capacitance (cap.): See table I.

Capacitance tolerance: ±10 percent (K), or ±20 percent (M).

Dissipation factor (DF): See table I.

Terminal strength: In accordance with method 211 of MIL-STD-202.

Test condition A: One-half pound for case sizes A and B, 1 pound for case sizes C and D, and 2 pounds for case sizes E, F, and G.

Test condition C: One-half pound applied force - 3 bends.

Stability at low and high temperature: In accordance with MIL-PRF-49137.

Step 1 (+25°C): DCL - See table I.
Cap. - Within tolerance specified in table I.
DF - See table I.

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Step 2 (-55°C):	ΔCap.	-	Within -15 percent of step 1 measured value.
	DF	-	200 percent of table I.
Step 3 (+25°C):	DCL	-	See table I.
	ΔCap.	-	Within 10 percent of step 1 measured value.
	DF	-	See table I.
Step 4 (+85°C):	DCL	-	10 times 25°C limit (see table I).
	ΔCap.	-	Within +15 percent of step 1 measured value.
	DF	-	150 percent of table I.
Step 5 (+25°C):	DCL	-	See table I.
	ΔCap.	-	Within 10 percent of step 1 measured value.
	DF	-	See table I.

Surge voltage: In accordance with MIL-PRF-49137.

DCL	-	See table I.
ΔCap.	-	Within 10 percent of initial value.
DF	-	See table I.

Life: Method 108 of MIL-STD-202.

1,000 hours:

At 25°C:	DCL	-	200 percent of table I.
	ΔCap.	-	Within 10 percent of initial measured value.
	DF	-	See table I.

Resistance to solvents: Not applicable.

Marking: In accordance with MIL-PRF-49137, except that case sizes A and B, shall be marked as shown in example 1. The front shall be marked with the capacitance value coded in picofarads and the reverse shall be marked with the dc voltage and the manufacturer's trademark. Case sizes C and D shall be marked as shown in example 2. The first line will contain the capacitance in picofarads and the capacitance tolerance. The second line will contain the dc voltage and the manufacturers trademark. Case sizes E, F, and G shall be marked as shown in example 3. The first line will contain the part number and the second shall contain the date code (year and week), lot code, and manufacturer's trademark.

At the option of the manufacturer, color coding may be used to mark the capacitance, dc rated voltage, and the manufacturers trademark on case sizes A and B. The use of two lines, the first line to identify the capacitance, and tolerance, and the second line, the dc voltage and manufacturers trademark for case sizes C and D. The use of three lines to mark the date, lot code, and manufacturers trademark on case sizes E, F, and G, as long as it is the same order as shown in the example.

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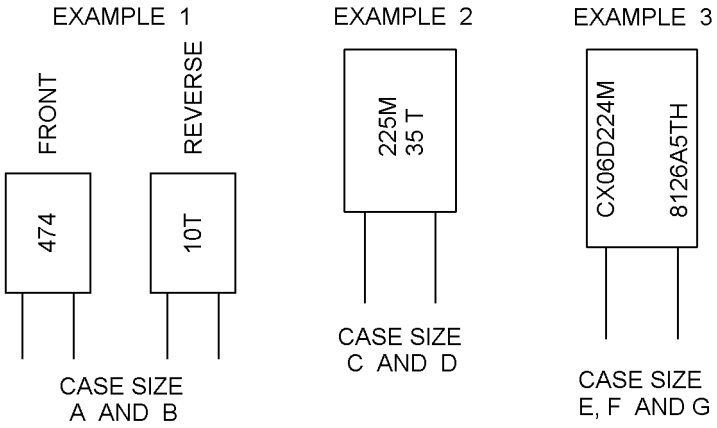

Voltage groups (85°C rated voltage):
I----------2 to 10 V inclusive
II----------15 to 35 V inclusive

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